Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/24/2005
PI Name:	Verghese, Preeti Ph.D.		
Project Title:	Presidential Award - Processing Motion Signals in Complex Environments - Phase II		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94115	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	09/30/2002	**End Date:**	09/23/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	An accurate representation of scene layout and of moving objects in the scene is required to navigate through complex environments. In the first part of the grant, our efforts were largely directed toward understanding how humans detect objects moving along a smooth path in a cluttered environment. More recently we have also looked at how depth information contributes to surface layout. . Our goal is to understand how local disparity and motion signals are combined to guide human locomotion and to provide information about depth, object boundaries and surface structure. Understanding how motion and depth information are combined in situations that include friction and gravity will help predict human perception under diverse conditions critical to NASA's mission of space exploration.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Earth Benefits: An accurate representation of scene layout and of moving objects in the scene is required to navigate through complex environments. Until recently our efforts were largely directed toward understanding how humans detect objects moving along a smooth path in a cluttered environment. Over the last year we have also begun to look at surfaces defined solely by disparity or depth information. Our goal is to understand how local disparity and motion signals are combined to guide human locomotion and to provide information about depth, object boundaries and surface structure.
Task Progress:	This past year was a no-cost extension of the grant primarily intended to write up results of studies completed in previous grant years (summarized in the Annual Report of January 2005). This goal has been accomplished---five papers have been accepted for publication this year in peer-reviewed journals (these papers are listed in the Bibliography section). I also worked with a post-doctoral fellow to initiate an investigation of the pattern of human eye movements when observers examine novel shapes. The results are summarized below: Human eye movements while viewing novel shapes. We examined human eye movements in the context of a general model for programming eye movements. The model reconstructs high-resolution visual information from a sequence of fixations, taking into account the fall-off in resolution from the fovea to the periphery. The model uses a simple rule for predicting fixation sequences: after each fixation, fixate next at the location that minimizes uncertainty (maximizes information) about the stimulus. We call this approach an information maximization model. We obtained human eye movement data from a separate study where humans were presented with novel shapes that they viewed peripherally for 300ms, then actively studied for 1.2 seconds while eye movements were measured with an SRI Dual Purkinje Image eye tracker. After the learning interval, the studied shape reappeared at a different size and location along with another shape that was very similar to the viewed shape. The observers' task was to select the shape they saw and rate their confidence. We compared these eye movement data to the predictions of our model as well as to other models that either choose "salient" locations or choose random locations as fixation targets. Our results demonstrate that our model is best at predicting fixation locations. Modeling additional biological constraints will improve the prediction of fixation sequences. Our results suggest that information maximization is a useful principle for programming eye movements.
Bibliography Type:	Description: (Last Updated: 03/15/2017)
Abstracts for Journals and Proceedings	Verghese, P.;, Petrov, Y.; McKee, S.P. (2004) "Collinear flankers improve detection by reducing uncertainty. " Society for Neuroscience San Diego CA. Society for Neuroscience Abstracts 2004 , Oct-2004
Abstracts for Journals and Proceedings	Renninger, L.W.; Verghese, P.; Coughlan, J.M. "Modeling eye movements in a shape discrimination task. " Vision Sciences Society, May 6 - 11, Sarasota, FL Vision Sciences Society Abstracts 2005 , May-2005
Abstracts for Journals and Proceedings	Verghese, P.; Petrov, Y.; McKee, S.P. "A role for uncertainty in collinear facilitation." Computational and Systems Neuroscience (COSYNE) 2005, Salt Lake City, UT COSYNE 2005 Abstracts , Mar-2005
Abstracts for Journals and Proceedings	Verghese, P.; Baldassi, S. "Selective modulation of attention with location and feature cues." American Psychological Society, Symposium on Visual Attention, Los Angeles CA, APS Abstracts. , May-2005
Articles in Peer-reviewed Journals	Vreven D.; Verghese P. "Predictability in the flash-lag effect. " Perception 34(1):31-44. , Jan-2005
Articles in Peer-reviewed Journals	Renninger, L.W.; Coughlan, J.;Verghese, P.; Malik, J. (2005) "An information maximization model of eye movements. " Advances in Neural Information Processing Systems 17: 1121-1128 , Jan-2005
Articles in Peer-reviewed Journals	Baldassi, S.; Verghese, P. " Attention to locations and features: different top-down modulation of detector weights." Journal of Vision 5(6): 556-70. , Jun-2005
Articles in Peer-reviewed Journals	Verghese P.; McKee S.P. "Motion grouping impairs speed discrimination." Vision Research (in press). , Sep-2005
Articles in Peer-reviewed Journals	McKee, S.P.; Verghese, P.; Farell B. "Stereo sensitivity depends on stereo matching" Journal of Vision (in press). , Dec-2005
Papers from Meeting Proceedings	Verghese, P. "Detecting motion trajectories and static contours in noise" Computation Vision in Neural and Machine Systems, Center for Vision Research, York University, Toronto, Canada, June 2005 Jun-2005